

Answers

1. When you multiply a vector with a scalar value, does the resulting vector change direction, magnitude, or both?

Answer: *Multiplying a vector by a scalar changes its magnitude. If the scalar is positive, the direction remains the same. If the scalar is negative, the vector reverses direction.*

2. Assume you have a direction vector V , how could you use vector scaling to ensure consistent movement in that direction every frame?

Answer: *So this was intentionally open ended, but basically you need to understand that you can scale the vector V by the time elapsed since the last frame. If the direction vector V is unit length, you can calculate the movement as $V * speed * timeElapsed$.*

3. Given 2 colours composed of RGB values, colour C and a tint colour T , how would you tint C by T ?

Answer: *You can simply multiply them together, element-wise, using the Hadamard product.*

4. Given a colour C composed of RGB values, and a scalar value of 0.5, what would happen if we multiplied them together? What is the effect on the colour?

Answer: *The basic idea here is that multiplying by values < 1 darkens the colour, so multiplying by 0.5 would halve the brightness.*

5. What if we did the reverse, and divided by 0.5 instead? What is the effect on the colour?

Answer: *The basic idea here is that multiplying by values > 1 brightens the colour. So dividing by 0.5 is the same as multiplying by 2, doubling the brightness.*

6. How could I scale a model's size uniformly using vector scaling?

Answer: *You would simply multiply every vertex position by the scalar value.*

7. What is element-wise multiplication of vectors called?

Answer: *The Hadamard product.*

8. What is element-wise multiplication of vectors useful for? Give some examples.

Answer: *2 major uses would be non-uniform scaling and colour tinting. In games, non-uniform scaling would be used to create stretching and compression effects on images and models, or to limit movement to certain axes for effects such as swaying grass. Colour tinting is useful in many different places, especially in graphics. Lighting relies heavily on tinting for light colour, and things like particle systems often change colour over time, which is another use for tinting.*

9. How would you flip the direction of a vector using vector scaling?

Answer: *Multiply by -1.*

10. Given $V = (3, 4)$, $S = 4$, what is SV ?

Answer: $(12, 16)$

11. Given $V = (1, 2)$, $S = -1.5$, what is SV ?

Answer: $(-1.5, -3)$

12. Given $V = (1, 2, 3)$, $S = 5$, what is SV ?

Answer: $(5, 10, 15)$

13. Given $V = (3, 4)$, $S = (1, 0)$, what is $S \circ V$?

Answer: $(3, 0)$

14. Given $V = (5, 6)$, $S = (5, 5)$, what is $S \circ V$?

Answer: $(25, 30)$

15. Given $V = (1, 2, 3)$, $S = (1, -1, 1)$, what is $S \circ V$?

Answer: $(1, -2, 3)$

16. Is scalar vector multiplication commutative?

Answer: Yes

17. Is it associative?

Answer: Yes

18. How about element-wise multiplication?

Answer: Both associative and commutative.